

- (a) $\alpha_{t_1 \rightarrow t_2}^{(l)} > 0.95$
- (b) $\alpha_{t_1 \rightarrow t_2}^{(l)} \in \text{top-3}(\{\alpha_{i \rightarrow t_2}^{(l)} : i \in [1, n]\})$: t_1 is among the top-3 attended-to tokens for t_2 at layer l
- (c) $\alpha_{t_1 \rightarrow t_2}^{(l)} \in \text{top-10}(\{\alpha_{t_1 \rightarrow j}^{(l)} : k \in [1, n]\})$: t_2 is among the top-10 attended-from tokens for t_1 at layer $l - 1$

We show example attention patterns for an AA model on three sample prompts in Figure 16. We only plot the attention subgraph directly connected to the final token position at the final layer (which is used to predict the state). We find that attention in AA models forms a tree-like pattern where successive layers attend to wider and wider context windows, with nodes that are more and more spaced apart. This is in line with how we believe the associative algorithm works: adjacent pairs of actions are grouped together at each layer in a hierarchical manner.

Note that the tree is not entirely clean: there are redundant edges and edges that cross over each other. We suspect that the subgraph we’ve picked up on is not a single tree, but rather the superimposition of *multiple* trees, each potentially contributing to not just the prediction for the final token, but also the predictions of the previous tokens.

G. How are these algorithms learned over the course of training?

We conduct a more detailed analysis of the training phases for two Pythia models trained on the S_5 task: one that learned AA and one that learned PAA. The training curves are shown in Figure 6, and we investigate the generalization behavior at different points along these curves. Both models improve over training by progressively generalizing to longer sequence lengths, rather than making uniform gains across all lengths. In the case of the PAA model, convergence appears to occur in two distinct phases: first, the model learns the parity of states across the entire length-100 sequence, followed by learning how to predict the state. By contrast, the AA model learns to generalize parity and state simultaneously.

H. Additional Factors Influencing Learned Algorithm

H.1. Model Size

We have investigated whether model size influences which algorithm the models learn to implement. As shown in Figure 17, model size empirically does not seem to have much effect on the choice of learned algorithm, with model architecture and initialization having a much bigger effect. Some notable differences between the GPT-2 and Pythia architecture are the use of rotary embeddings, parallelized attention, and feedforward layers rather than sequential, and untied embedding and unembedding.

H.2. Topic Modeling

The topic model used in Section 5.3 is parameterized as follows: We generate the distribution of the 4 topics in each document using a random Dirichlet distribution with $\alpha = 0.3$. The distribution $p(\text{token} \mid \text{topic})$ for each token 123, 132, 213, 231, 312, 321 is:

$$\begin{bmatrix} 3.06 \cdot 10^{-2}, & 1.11 \cdot 10^{-1}, & 5.79 \cdot 10^{-4}, & 6.45 \cdot 10^{-3}, & 6.58 \cdot 10^{-3}, & 8.45 \cdot 10^{-1} \\ 3.36 \cdot 10^{-1}, & 1.69 \cdot 10^{-4}, & 6.63 \cdot 10^{-1}, & 8.05 \cdot 10^{-7}, & 1.68 \cdot 10^{-7}, & 7.81 \cdot 10^{-4} \\ 7.92 \cdot 10^{-5}, & 1.41 \cdot 10^{-2}, & 9.44 \cdot 10^{-1}, & 4.53 \cdot 10^{-4}, & 4.13 \cdot 10^{-2}, & 3.27 \cdot 10^{-11} \\ 2.85 \cdot 10^{-3}, & 1.29 \cdot 10^{-9}, & 7.06 \cdot 10^{-1}, & 6.37 \cdot 10^{-7}, & 2.58 \cdot 10^{-3}, & 2.89 \cdot 10^{-1} \end{bmatrix}$$

We also trained LMs using a topic model with a second token-topic distribution, aiming to distinguish the effect of this particular topic distribution from the effect of topic modeling pretraining in general. On the second distribution, we also find that both randomly initialized GPT-2 and Pythia models learn AA in Figure 18. The $p(\text{token} \mid \text{topic})$ distribution for this model is listed below:

$$\begin{bmatrix} 9.31 \cdot 10^{-1} & 2.32 \cdot 10^{-3} & 1.38 \cdot 10^{-8} & 5.86 \cdot 10^{-10} & 4.62 \cdot 10^{-5} & 6.63 \cdot 10^{-2} \\ 2.10 \cdot 10^{-4} & 3.12 \cdot 10^{-4} & 9.32 \cdot 10^{-7} & 9.07 \cdot 10^{-6} & 2.53 \cdot 10^{-1} & 7.47 \cdot 10^{-1} \\ 4.95 \cdot 10^{-1} & 1.18 \cdot 10^{-3} & 4.55 \cdot 10^{-1} & 2.17 \cdot 10^{-2} & 1.86 \cdot 10^{-8} & 2.71 \cdot 10^{-2} \\ 6.55 \cdot 10^{-1} & 4.92 \cdot 10^{-4} & 3.44 \cdot 10^{-1} & 2.28 \cdot 10^{-7} & 1.94 \cdot 10^{-4} & 2.14 \cdot 10^{-8} \end{bmatrix}$$

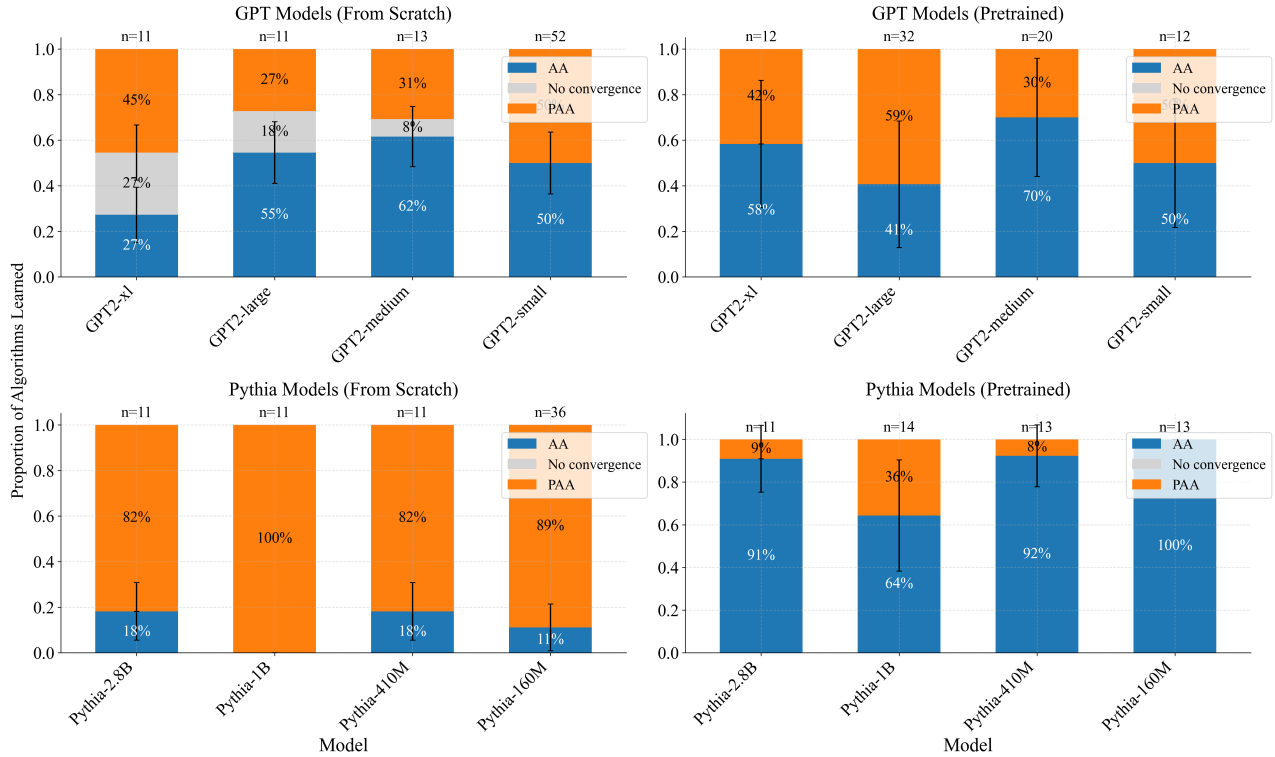


Figure 17. Proportion of S_3 algorithms learned by models of various sizes from the GPT-2 and Pythia families. Model architecture and initialization are much bigger factors in influencing the algorithms learned than model size.

H.3. Parity Loss Curriculum

We explored an additional procedure that encourages models to learn parity via an extra loss term. We train an extra linear classifier that takes in the residual activations of an early layer (e.g., layer 3) in which we find parity to be typically computed through probing. We train with this additional loss term to induce the representation to linearly encode parity early on. The classifier is trained to output (1) parity or (2) a (parity, action) tuple, to ensure that the residual also encodes the original action, and not just the parity. After training, we evaluate the model on the original S_3 task. Both procedures induce the model to learn PAA consistently, though the (parity, action) classifier typically allows the model to generalize better.

H.4. Length Curriculum

We implemented a curriculum training approach where the model was progressively exposed to documents of increasing length. We first trained on only the initial 10 tokens, then expanded to 25 tokens, 50 tokens, and finally the complete 100-token sequences. Each stage of the curriculum was trained for a fixed number of epochs (data). The goal of such a curriculum is to push the model to learn the associative algorithm (AA), as the parity heuristic, we hypothesize, might be less useful for shorter sequences. However, empirically, such a curriculum has no obvious effect on the kind of algorithm the model learns. Of the 5 trials, 2 trials of GPT-2 learn AA, and the other 3 trials learn PAA.

We discovered that all five models can perfectly generalize both parity and state when trained on the first 25 tokens. It is only when trained to generalize to sequence length 50 that the distinction between PAA and AA emerges. Models learning the PAA algorithm do not generalize, while models learning AA can generalize from length 25 to 50, as illustrated by Figure 19. This further confirms our finding that the model learns the S_3 algorithm early on.

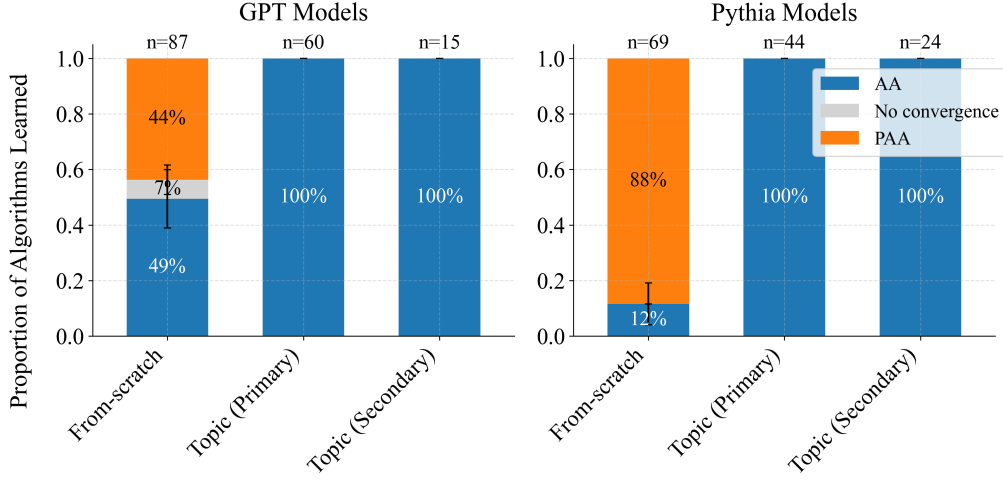


Figure 18. Proportion of S_3 algorithms learned by models first pre-trained on the two topic models compared to the randomly initialized baselines. Topic modeling, regardless of the specific distributions, pushes the model to learn AA.

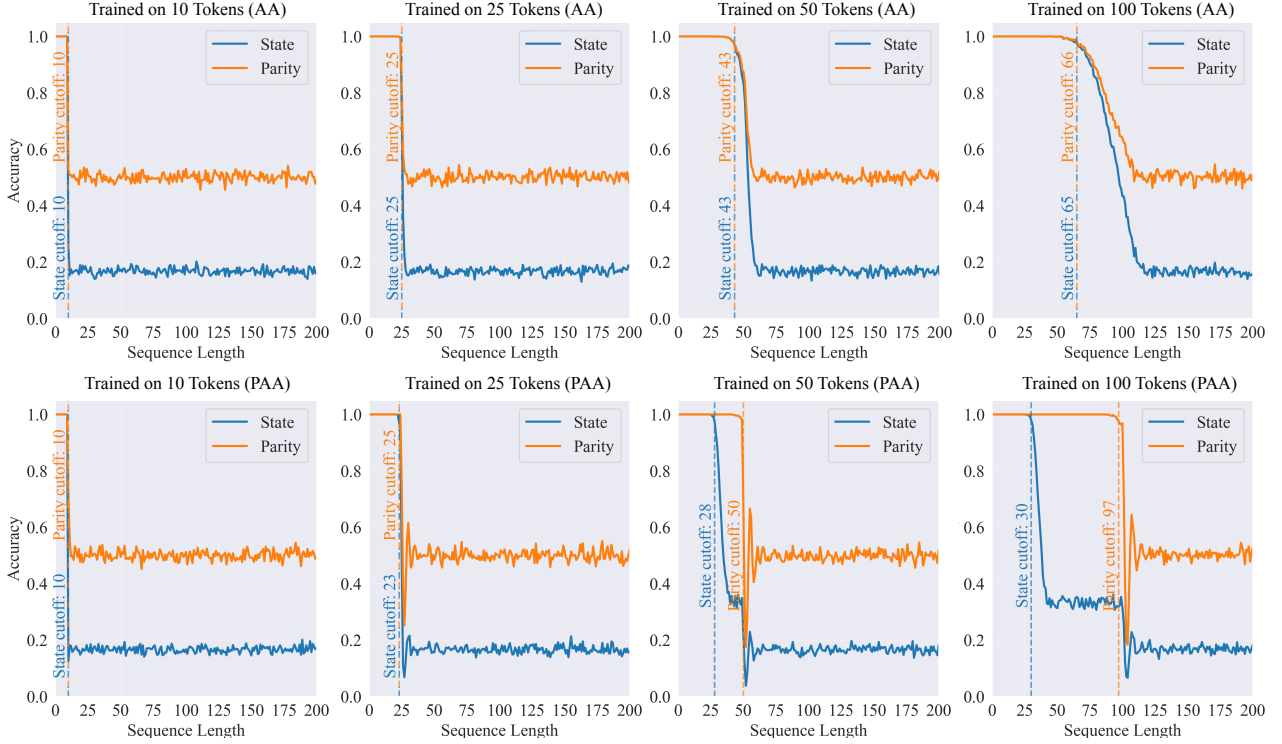


Figure 19. Model generalization curves when trained using a length curriculum after training on 10, 25, 50, and 100 tokens, respectively. While the length curriculum doesn't push the model to learn one algorithm or the other, it shows that the model learns these algorithms early on, as indicated by whether the model can generalize well from training on 25 tokens to 50 tokens.